

Repository Breakdown

Complete structural overview of the Rental Inventory & Booking Management System.

Tech Stack

Layer	Technology
Framework	Next.js 16.2.7 (App Router, Turbopack)
Language	TypeScript
Styling	Tailwind CSS v4 + tw-animate-css + shadcn/tailwind.css
UI Components	shadcn/ui (Base UI primitives)
Fonts	Inter (body) + Outfit (headings) via next/font/google
Database	PostgreSQL via Supabase
Auth	Supabase Auth (email/password, role-based)
Forms	React Hook Form + Zod validation
Calendar	react-day-picker
Icons	lucide-react
Toasts	Sonner
Charts	Recharts
Email	Resend (optional)
Hosting	Vercel
Package Manager	npm

Root Files

— .env.local	# Local environment variables (Supabase keys, app URL)
— .env.local.example	# Template for .env.local
— .env.local.vercel	# Vercel-specific env template
— .env.example	# General env template
— .gitignore	
— components.json	# shadcn/ui configuration
— eslint.config.mjs	# ESLint flat config
— middleware.ts	# Next.js middleware – delegates to Supabase session refresh
— next.config.ts	# Next.js configuration
— next-env.d.ts	# TypeScript declarations for Next.js

```
|— package.json           # Dependencies and scripts
|— package-lock.json
|— postcss.config.mjs     # PostCSS (Tailwind plugin)
|— tsconfig.json          # TypeScript config
|— tsconfig.tsbuildinfo   # Build cache
```

middleware.ts

Runs on every request. Delegates to `lib/supabase/middleware.ts` which refreshes the Supabase session cookie and protects `/admin` routes (redirects unauthenticated users to `/login`).

app/ — Next.js App Router Pages

Layout & Global

File	Type	Purpose
layout.tsx	Root layout	Loads Inter + Outfit fonts, renders <code><Toaster></code> , wraps all pages
globals.css	Global styles	Tailwind imports, <code>@theme inline</code> CSS variables, animation keyframes (<code>fade-in</code> , <code>float</code> , <code>shimmer</code> , etc.), utility classes (<code>.heading</code> , <code>.glass-card</code> , <code>.hover-lift</code> , <code>.animate-gradient</code>)
page.tsx	/ — Home page	Server component. Dark hero with animated gradient, floating orbs, CTA buttons, "Why Choose Us" features, featured items grid, contact section, footer
favicon.ico	Favicon	Browser tab icon

app/login/

File	Purpose
page.tsx	Admin login page. Client component. Gradient background with animated blurred circles, frosted glass card, email/password form with Zod validation, show/hide password toggle

app/admin/ — Admin Dashboard

All admin pages are **server components** that check auth via `supabase.auth.getUser()` and redirect to `/login` if not authenticated. Only users with role `owner`, `admin`, or `staff` can access.

File	Purpose
layout.tsx	Admin shell. Checks auth + role, fetches business name. Sticky header with floating Package icon, sidebar navigation (Dashboard, Inventory, Bookings, Calendar, Settings), mobile bottom nav bar
page.tsx	Dashboard overview. Stat cards (pending bookings, confirmed, total revenue, cancelled, total items) with gradient icons and hover lift. Recent

File	Purpose
	bookings list. Quick actions. "At a Glance" gradient card
<code>bookings/page.tsx</code>	Bookings list. Table with customer name, email, phone, event dates, status badges (color-coded), total amount. Filters by search and status. OR filter for public bookings
<code>bookings/[id]/page.tsx</code>	Single booking detail. Client component. Full booking info, items list, status update dropdown, admin notes textarea, confirm/cancel/complete actions with confirmation dialog
<code>inventory/page.tsx</code>	Inventory list. Table with item name, category pills, price, total quantity, active status. Search by name
<code>inventory/new/page.tsx</code>	Create new item form. Name, category, description, price, quantity fields
<code>inventory/[id]/page.tsx</code>	Edit existing item. Same form as create, pre-filled
<code>calendar/page.tsx</code>	Calendar view (placeholder)
<code>settings/page.tsx</code>	Settings page (placeholder)

`app/booking/` — Public Booking Flow

File	Purpose
<code>page.tsx</code>	Multi-step booking form (client component). Step 1: Date selection with dual date pickers (react-day-picker in popovers). Step 2: Item selection with availability check, quantity +/- controls, category grouping, gradient cart summary. Step 3: Customer info form (name, phone, email, address, event type, notes) + dark booking summary panel. Submits to <code>POST /api/bookings</code>
<code>confirmation/page.tsx</code>	Booking confirmation. Server component. Fetches booking by ID from URL params. Dark hero with animated check icon, booking details card, items list with totals, next steps info

`app/catalog/` — Public Item Catalog

File	Purpose
<code>page.tsx</code>	Server component. Dark hero banner. Search + category filter form. Items grouped by category with colored accent bars, hover-lift cards showing name, price, description, availability, "Book" link

`app/business/[slug]/` — Business-Specific Booking

File	Purpose
<code>page.tsx</code>	Business landing page. Fetches business by slug, shows items

File	Purpose
booking-form.tsx	Business-specific booking form. Similar to /booking but scoped to a specific business's inventory
book/page.tsx	Alternative booking page for a business

app/api/ — API Routes

Endpoint	Methods	Purpose
bookings/route.ts	GET, POST	POST: Create public booking. Validates fields, calculates <code>total_amount</code> , uses service client (bypasses RLS) for inserts, sends emails (non-blocking). GET: Fetch booking by ID for confirmation page
bookings/[id]/route.ts	PATCH, DELETE	PATCH: Update booking status (pending → confirmed → completed/cancelled). Rechecks availability on confirm. Sends status change emails. DELETE: Remove a booking
items/route.ts	GET, POST	GET: List all active items. POST: Create new item (admin)
items/[id]/route.ts	GET, PATCH, DELETE	GET: Single item. PATCH: Update item. DELETE: Remove item
availability/route.ts	GET	Returns item availability for a date range. Calculates available quantity = total - booked for overlapping dates
business/route.ts	GET, POST	GET: Fetch business by slug. POST: Create/update business
auth/signout/route.ts	POST	Signs out the current user and redirects to /

components/ — UI Components

components/ui/ — shadcn/ui Components

All are reusable, styled UI primitives built on Base UI or Radix.

File	Component	Notes
badge.tsx	Badge	Status/label badges with variants
button.tsx	Button, buttonVariants	Primary UI button. All variants have <code>hover:scale-[1.02]</code> , <code>active:scale-[0.98]</code> , <code>hover:cursor-pointer</code>
calendar.tsx	Calendar	react-day-picker wrapper. White background, violet selected state with scale animation, rounded-xl
card.tsx	Card, CardHeader, CardContent, etc.	Standard card layout components
dialog.tsx	Dialog	Modal dialog (Base UI)

File	Component	Notes
dropdown-menu.tsx	DropdownMenu	Dropdown menus
form.tsx	Form, FormField, FormControl, etc.	React Hook Form integration components
input.tsx	Input	Text input field
label.tsx	Label	Form label
popover.tsx	Popover, PopoverTrigger, PopoverContent	Popover primitive. Critical: <code>PopoverTrigger</code> renders as native <code><button></code> — do NOT nest <code><Button></code> inside
select.tsx	Select, SelectTrigger, etc.	Dropdown select
separator.tsx	Separator	Visual divider
skeleton.tsx	Skeleton	Loading placeholder
sonner.tsx	Toaster	Toast notification provider
table.tsx	Table, TableHeader, TableRow, etc.	Data table layout
textarea.tsx	Textarea	Multi-line text input

lib/ — Business Logic & Utilities

lib/supabase/ — Supabase Client Configuration

File	Purpose	Key Details
server.ts	Server-side client	Uses <code>@supabase/ssr</code> + <code>cookies()</code> . Prefers <code>NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY</code> , falls back to <code>NEXT_PUBLIC_SUPABASE_ANON_KEY</code> . Used by server components and API routes
client.ts	Browser-side client	Uses <code>@supabase/ssr createBrowserClient</code> . Same key preference. Used by client components (login page, booking detail)
middleware.ts	Session refresh + auth guard	Creates client with request/response cookies. Calls <code>supabase.auth.getUser()</code> . Protects <code>/admin/*</code> routes — redirects to <code>/login</code> if no user
service-client.ts	Admin client (bypasses RLS)	Uses <code>@supabase/supabase-js</code> directly (NOT the SSR wrapper). No cookie auth — pure <code>service_role</code> key. Used for public booking creation to avoid RLS conflicts

lib/services/ — Business Logic

File	Purpose
availability.ts	validateBookingAvailability() and getItemAvailability(). Calculates how many of each item are available for a given date range by subtracting booked quantities from total. Supports filtering by categories, item IDs, statuses, and excluding a specific booking
availability-examples.ts	Usage examples for the availability service
email.ts	Email notification functions using Resend. sendCustomerBookingRequest(), sendAdminNewBookingNotification(), sendCustomerBookingConfirmed(), sendCustomerBookingCancelled(), sendCustomerBookingCompleted(). All wrapped in try-catch — silent fail if RESEND_API_KEY isn't set

lib/types/

File	Purpose
database.types.ts	Auto-generated Supabase TypeScript types. Defines Item, Booking, Profile, Business, BookingItem table types and their relationships

lib/utils.ts

cn() helper — merges Tailwind classes using clsx + tailwind-merge.

supabase/ — Database

Migrations

File	Purpose
001_initial_schema.sql	Creates core tables: profiles (extends auth.users with role), items (rental inventory), bookings (customer reservations), booking_items (junction table). Enables RLS with basic policies
002_multi_tenant.sql	Adds businesses table, business_id foreign keys to profiles/items/bookings. Multi-tenant RLS policies using helper functions (get_user_business_id(), user_has_role()). Seeds a default business. Creates profile on user signup via trigger
003_public_bookings_rls.sql	Adds RLS policies for public bookings (where business_id IS NULL). Lets authenticated admins SELECT and UPDATE unassigned bookings and SELECT their booking items

Tables

businesses	– Multi-tenant business entities (name, slug, email, etc.)
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├─ profiles      - User accounts (extends auth.users). Has role
                    (owner/admin/staff/customer)
├─ items          - Rental inventory (name, category, price, quantity, active
                    status)
├─ bookings       - Customer reservations (customer info, dates, status,
                    total_amount)
├─ ┌─ booking_items - Junction: which items, how many, at what price
└─ └─ (items, bookings scoped to a business via business_id)
```

public/ — Static Assets

Default Next.js SVG assets (not yet customized):

- [file.svg](#), [globe.svg](#), [next.svg](#), [vercel.svg](#), [window.svg](#)

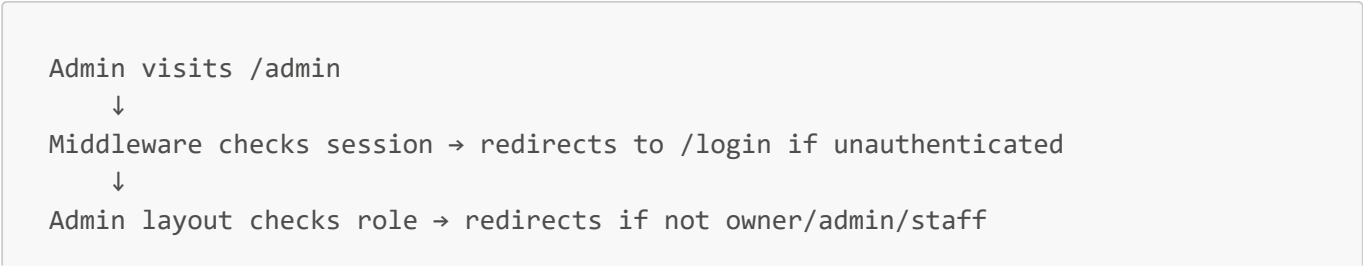
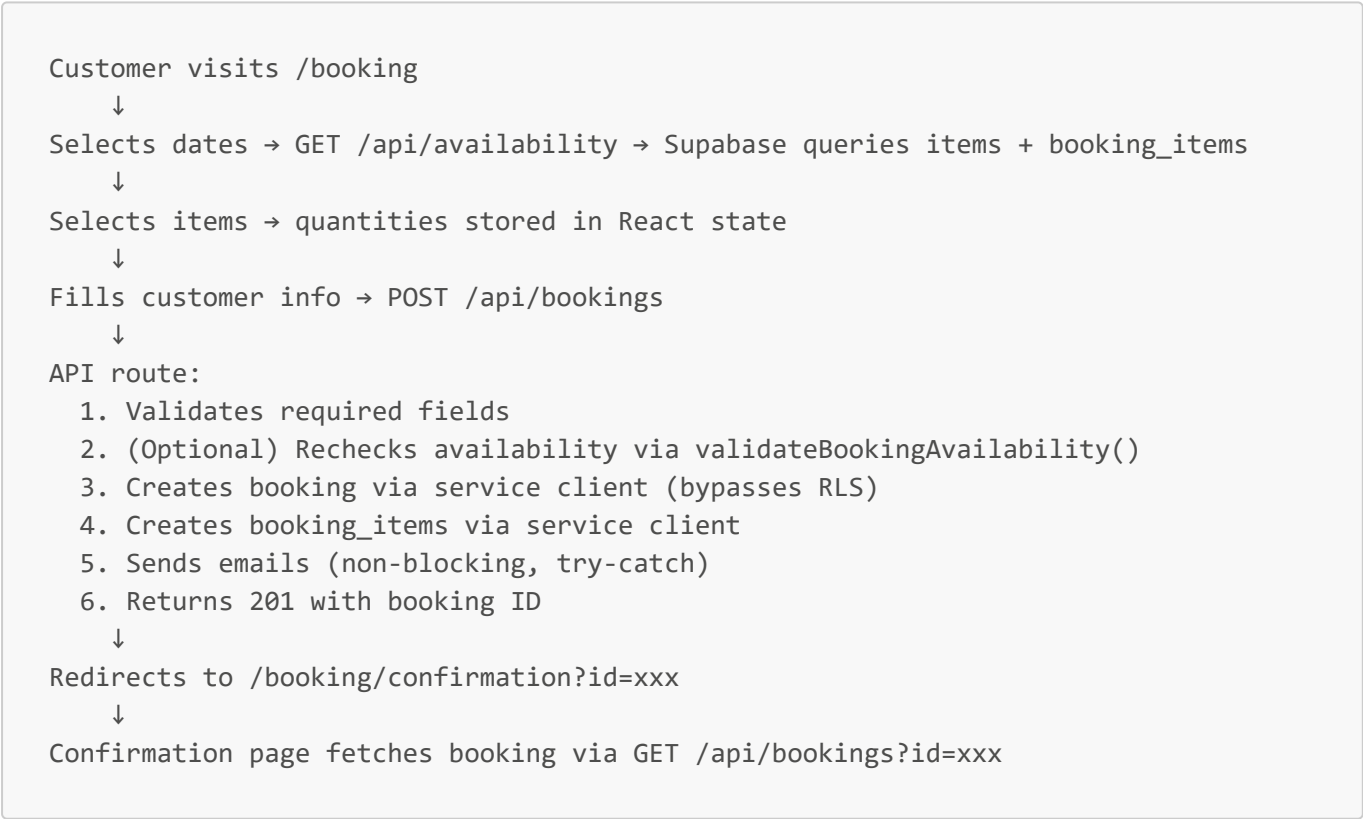
Documentation Files

File	Purpose
README.md	Project overview, feature list, tech stack, getting started
DEPLOYMENT.md	Step-by-step deployment guide for Supabase + Vercel
TROUBLESHOOTING.md	12 common issues with symptoms, cause, and fix
TESTING_GUIDE.md	Manual test scenarios for the booking flow
QUICKSTART.md	Quick setup instructions
PROJECT_OVERVIEW.md	High-level project description
ADMIN_DASHBOARD.md	Admin dashboard documentation
BOOKING_FLOW.md	Booking flow architecture
AVAILABILITY_SERVICE.md	Availability calculation logic
AVAILABILITY_ENHANCEMENTS.md	Availability feature improvements
CALENDAR_FEATURES.md	Calendar feature docs
STATUS.md	Project status tracking
build_plan.md	Build plan / roadmap
AGENTS.md	AI agent rules (Next.js-specific)
CLAUDE.md	References AGENTS.md

Environment Variables

Variable	Required	Where Used	Purpose
NEXT_PUBLIC_SUPABASE_URL	✓	All Supabase clients	Project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	✓	server.ts, client.ts, middleware.ts	Anonymous key (fallback)
NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY	Recommended	server.ts, client.ts, middleware.ts	New publishable key (preferred over anon)
SUPABASE_SERVICE_ROLE_KEY	✓	service-client.ts	Admin key — bypasses RLS for booking creation
NEXT_PUBLIC_APP_URL	Optional	Confirmation page	Public URL for email links
RESEND_API_KEY	Optional	email.ts	Email sending via Resend

Data Flow





Dashboard queries bookings + items → displays stats
Bookings page lists all bookings (business-scoped + public/unassigned)

Animation System

Defined in `globals.css`:

Animation	Usage
<code>animate-fade-in</code>	General entrance
<code>animate-fade-in-up</code>	Cards, sections rising into view
<code>animate-fade-in-down</code>	Header/nav sliding down
<code>animate-slide-in-right</code>	Side panels
<code>animate-scale-in</code>	Modals, popovers
<code>animate-float</code>	Decorative icons, orbs
<code>animate-pulse-soft</code>	Status indicators
<code>animate-shimmer</code>	Loading states
<code>stagger-children</code>	Sequential entrance for lists (nth-child delays)
<code>animate-gradient</code>	Hero background gradient shift
<code>hover-lift</code>	Card hover → translateY(-4px) + shadow
<code>gradient-text</code>	Gradient clipped to text
<code>glass-card</code>	Frosted glass effect